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PRINCIPAL INTERESTS	Computer Vision, Omnidirectional Cameras, Dual-Pixel Cameras, Vision-based Depth Estimation
ACADEMIC BACKGROUND	<i>Ph.D. Mechanical Engineering</i> Aug 2020 - KAIST, Daejeon, South Korea - Focus Areas: Computer Vision, Occupancy Prediction, Dual-Pixel Sensor <i>M.S. Mechanical Engineering</i> Aug 2018- Aug 2020 KAIST, Daejeon, South Korea - Focus areas: Computer Vision. Omnidirectional Vision / Depth Estimation <i>B.S. Mechanical Engineering</i> Aug 2014- May 2018 University of Illinois at Urbana-Champaign, Champaign, IL - Minor in Computer Science
EMPLOYMENT HISTORY	<i>Graduate Researcher</i> Aug 2018 - Present KAIST, VILab, Daejeon, South Korea - Project Lead/Manager for Several Govt./Industry-funded Projects - Research on <i>Depth Estimation using Dual-Pixel Camera</i> <i>Undergraduate Research Assistant</i> Spring 2017 - Winter 2017 UIUC, Dynamic Robotics Lab, Champaign, IL - Designed Test benches for testing portions of quadrupedal robot - Assisted graduate student by improving assembly processes and testing processes through critical input - When assigned with unfamiliar tasks, reached out to available resources to become adept at the given task <i>Undergraduate Research Assistant</i> Summer 2016 - Spring 2017 UIUC, John Rogers Research Group, Champaign, IL - Fast paced research group that required strict deadlines regarding progress - Worked remotely with a graduate student on a project regarding manufacturing of MEMS device - Cooperated with graduate student on project by adding critical insight as to improve manufacturing process <i>Course Assistant</i> Spring 2016 TAM212 Dynamics Course, UIUC, Champaign, IL - Fast paced research group that required strict deadlines regarding progress - Worked remotely with a graduate student on an ongoing project regarding manufacturing of MEMS device - Cooperated with graduate student on project by adding critical insight as to improve manufacturing process

**Publications
(International)**

SCI Journal

6. Cho, I., Lee, K., Sim, Y. C., **Jeong, J. S.**, Cho, M., Jung, H., Kang, M., Cho, Y. H., Ha, S. C., Yoon, K. J., & Park, I. (2025). Correction: Deep-learning-based gas identification by time-variant illumination of a single micro-LED-embedded gas sensor. *Light, science & applications*, 14(1), 380. <https://doi.org/10.1038/s41377-025-02023-5>
5. Cho, I., Lee, K., Sim, Y. C., **Jeong, J. S.**, Cho, M., Jung, H., ... & Park, I. (2023). Deep-learning-based gas identification by time-variant illumination of a single micro-LED-embedded gas sensor. *Light: Science & Applications*, 12(1), 95.
4. Han, J. K., Kang, M., **Jeong, J.**, Cho, I., Yu, J. M., Yoon, K. J., ... & Choi, Y. K. (2022). Artificial Olfactory Neuron for an In-Sensor Neuromorphic Nose. *Advanced Science*, 2106017.
3. Kang, M., Cho, I., Park, J., **Jeong, J.**, Lee, K., Lee, B., ... & Park, I. (2022). High Accuracy Real-Time Multi-Gas Identification by a Batch-Uniform Gas Sensor Array and Deep Learning Algorithm. *ACS sensors*.
2. Cho, Hoonhee, **Jaeseok Jeong**, and Kuk-Jin Yoon. "EOMVS: Event-Based Omnidirectional Multi-View Stereo." *IEEE Robotics and Automation Letters* (2021)
1. Lee, Yeonkun, **Jaeseok Jeong**, Jongseob Yun, Wonjun Cho, and Kuk-Jin Yoon. "SpherePHD: Applying CNNs on 360° Images with Non-Euclidean Spherical PolyHeDron Representation." *IEEE Transactions on Pattern Analysis and Machine Intelligence* (2020).

Conference

6. Jeong, J., Yang, H., **Jeong, J.**, & Yoon, K. J. (2025). Improving Black-Box Generative Attacks via Generator Semantic Consistency . *under review*.
5. Yoon, S. H., Kwon, H., **Jeong, J.**, Park, D., & Yoon, K. J. (2024, September). Diffusion-guided weakly supervised semantic segmentation. In *European Conference on Computer Vision* (pp. 393-411). Cham: Springer Nature Switzerland.
4. Kwon, H., **Jeong, J.**, Yoon, S. H., & Yoon, K. J. (2024, September). Phase concentration and shortcut suppression for weakly supervised semantic segmentation. In *European Conference on Computer Vision* (pp. 293-312). Cham: Springer Nature Switzerland.
3. Kim, T., **Jeong, J.**, Cho, H., Jeong, Y., & Yoon, K. J. (2024, September). Towards real-world event-guided low-light video enhancement and deblurring. In *European Conference on Computer Vision* (pp. 433-451). Cham: Springer Nature Switzerland.
2. Park, D., **Jeong, J.**, Yoon, S. H., Jeong, J., & Yoon, K. J. (2024). T4p: Test-time training of trajectory prediction via masked autoencoder and actor-specific token memory. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition* (pp. 15065-15076).
1. Lee, Yeonkun*, **Jaeseok Jeong***, Jongseob Yun*, Wonjun Cho, and Kuk-Jin Yoon. "SpherePHD: Applying CNNs on a Spherical PolyHeDron Representation of 360° Images" *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition* (2019).

Patents

Domestic

2. 권혁준, **정재석**, 박휘성, 윤국진, "다시점 영상 스티칭을 위한 색 보정 방법, 영상 합성 방법, 영상 처리 장치, 저장 매체 및 컴퓨터 프로그램," KR 등록번호 10-2021-0167053
1. 이연건, **정재석**, 윤종섭, 조원준, 윤국진, "전방향 영상의 딥러닝을 이용한 물체 감지 방법 및 장치, 그리고 이를 이용한 차량 자율 주행 방법 및 드론 자율 주행 방법," KR 등록번호 10-2020-0048643

Awards

3. IPIU 2022 우수논문상 금상: Exploring Pixel-level Self-supervision for Weakly Supervised Semantic Segmentation
2. 1st Place in the Event-Only Track in DSEC Challenge: A Stereo Event Camera Dataset for Driving Scenario, CVPRW 2021 Event-Based Vision
1. IPIU 2019 우수논문상 금상: 정20면체 기반 360도 이미지 표현 및 CNN 적용 방법

Research Projects

8. X-Ray Denoising: DK Medical Systems, 08/2025 -
7. Intelligent AutoFocus Technology using Dual-Pixel Sensor: Samsung Electronics DS, 09/2021 - 09/2025
6. Development of Obstacle Height Estimation in Naval Operation Videos: Samsung Heavy Industries, 06/2022 - 07/2022
5. Learning-based Metallic Surface Grade Classification: Samsung Heavy Industries, 09/2021 - 12/2021
4. Perception for Collision Avoidance and Accident Prevention in Autonomous Naval System: KSOE, 09/2021 - 02/2022
3. Development of Quadruiped Robot for Surveillance, Reconnaissance, and Search Missions: ADD, 12/2019 - 11/2021
2. Development of Mobile Ground Station for Unmanned Swarm Cyber Physical System (CPS): ADD, 11/2019 - 03/2021
1. AAVM Pedestrian Detection: Hyundai Construction Equipment, 02/2019 - 12/2019